Deliverable 2 — Direction & Early Results

Sephora Customer Segmentation · Milan Viallet · April 2, 2026

1. A. Retrospection — Since Grill 1

Three feedback items reshaped the project since Grill 1: **(1)** *"PCA kills interpretability"* ⇒ removed PCA from clustering; we now cluster on **46 scaled features** directly (PCA kept only for correlation circle visualization). **(2)** *"Too many noisy features"* ⇒ classified features into 6 marketing categories, removed 16 non-actionable ones (70 → 46 features). **(3)** *"Track experiments"* ⇒ MLflow integration planned next sprint. **Scope:** depth-first K-Means with Hierarchical + GMM as comparison baselines.

2. B. Analysis — Method & Results

Method. K-Means on n=64,469 customers, 46-dim feature space. Pipeline: transactions → customer aggregation (RFM + channel + product + behavior) → drop non-actionable → median imputation + OHE → StandardScaler → K-Means (k=2–30). k=9 via elbow + silhouette (0.156) + Davies–Bouldin (1.63). Modest silhouette typical for retail behavioral data (0.10–0.25 in literature).

Figure 1: *

Top: KPI deviation (%) vs. global avg. **Bottom:** Product affinity (L) & channel mix (R) per cluster.

2.1 Clusters vs. Global Average

C	%	Monet.	Freq	Rec.	Bask.	Disc.	Key Insight
All	*100*	*€198*	*3.0*	*111d*	*€70*	*12%*	*Global baseline*
0	30.7	€114	2.7	117d	€43	6%	Low-spend Make Up store-only
1	10.6	€189	1.8	128d	€122	20%	High-basket Fragrance gifters
2	5.9	€1,010	13.4	24d	€85	16%	**VIP loyalists (49% Gold)**
3	8.4	€148	2.0	120d	€80	20%	Digital-first (95% e-store)
5	19.9	€223	2.6	101d	€95	20%	Mid-value Fragrance fans
6	20.7	€100	2.1	130d	€50	5%	Dormant low-engagement
7	2.9	€125	2.2	90d	€61	0%	Others-axis, zero discount
8	1.0	€273	3.4	108d	€79	9%	High-freq multi-brand

C2 (5.9%) = **5.1**× global monetary, 49% Gold. C3 confirms digital-native hypothesis (95% e-store ≠ 80% store base). C4 (4 customers) excluded as noise.

3. C. Evaluation & Confidence

Done: **(1)** Internal metrics — Silhouette = 0.156, DB = 1.63, CH = 6,471; **(2)** UMAP 2D shows clusters in visually distinct regions; **(3)** Business coherence — every cluster maps to a plausible marketing narrative (table above); **(4)** Correlation circle confirms no remaining redundant pairs (cos >0.90) after feature refonte.

Planned: **(1)** Kruskal–Wallis tests on all KPIs for statistical significance; **(2)** Bootstrap stability (n=50, ARI > 0.80 target); **(3)** Agglomerative + GMM comparison to rule out algorithm artifacts.

Weaknesses: **(1)** Cluster 4 (4 customers) = micro-cluster artifact. Will absorb or flag. **(2)** Age imputed with median for 13% missing — may flatten age-based differentiation. Fix: `IterativeImputer` with RF.

4. D. Roadmap — Path to Finish

#	Task	Description	ETA
1	Pipeline cleanup	Remove PCA, integrate MLflow tracking	Apr 3
2	Notebook refonte	End-to-end clean notebook	Apr 4
3	Algorithm comparison	Hierarchical, GMM; compare metrics	Apr 5
4	Validation	Kruskal–Wallis + bootstrap stability	Apr 6
5	Personas + Impact	Name segments, persona cards, CLV delta, prioritization matrix	Apr 7–8
6	Final delivery	Polished PDF + code + recommendations	Apr 9

Business value: CLV gap per segment → estimate revenue uplift from targeted campaigns. Example: moving C6's basket from €50 to €70 (+40%) across 13,368 customers × 2.1 visits = **€560K incremental annual revenue**.